

MAKING ELECTRONICS MANUFACTURING Predictably Profitable USING DATA ANALYSIS

Electronics manufacturers sitting on untapped data goldmines can drive revenue growth by breaking down operational silos and applying strategic analytics frameworks.



Each operation generates a large volume of data.

The solution is not a revolutionary technology; it is simply about connecting previously isolated data streams and analysing air temperature, water temperature, and pipe conditions alongside production outcomes. Over time, this analytical framework generates value, suggesting total returns exceeding half a billion dollars.

Companies sitting on years of unused historical data, quality metrics, and operational information can, when properly analysed, reveal optimisation opportunities worth millions.

The five-step ROI framework every decision maker needs

To implement data analysis successfully, investments must follow a predictable pattern across electronics manufacturing:

Scenario creation. Quantify specific problems by analysing costs, production inefficiencies, and quality failures, with precise financial impact measurements.

Data integration. Connect all isolated systems (SCADA, MES, PLC, historian databases) that currently operate as information silos.

SCADA provides a real-time view of pumps, valves, and temperature sensors; MES tracks every production order and quality check as it happens; PLCs drive machine logic, mixing, conveying, and heating; while historian databases archive all data points with timestamps. Pulling these together turns isolated signals into

While sensors generate terabytes of data from production lines and quality systems that store data of good and defective material, most organisations cannot transform this scattered data into usable information for strategic advantage, leaving profits on the table that competitors are beginning to claim.

Why invest in analytical data?

Large manufacturing enterprises run complex, end-to-end operations—from raw-material sourcing and processing plants to warehouse storage, distribution centres, and logistics partners.



ARVIND KUMAR PRASAD is Founder & Director of Vortexa Technology Advisors